

ZĖ-R0 Instrument Baseline Report v0.1 (Orthography Aperture Meter)

What this is

A reproducible baseline report for ZĖ-R0's **orthography aperture meter** and **baseline-lock** (drift) workflow.

This is a **measurement + reproducibility** report. It does not claim "meaning proof" by itself.

Method (SSOT + meter)

SSOT

- Orthography carrier extraction:

`src/shared/vowels/extractOrthographyVoicesFromWord.v0.1.ts`

Aperture proxy (fixed calibration scale)

- A=1.0, O=0.8, E=0.6, Ė=0.5, U=0.4, Y=0.3, I=0.1

Readouts

- **primary** = aperture of the first carrier

- **presence mean** = mean aperture over unique carriers (in order)

Slope test

- Pearson r and Spearman ρ on **bucket means** vs semantic index 1..7

- Permutation p-value: 12,000 iterations; fixed seed

Baseline-lock workflow

Harnesses always write **current** artifacts to:

- `tests/validation/out/\*.md|json`

Baselines are written to:

- `tests/validation/baselines/\*.md|json`

...and only when the dataset reaches its target per-bucket N (to avoid churn).

Baseline-locked datasets

Positive-signal:

- Albanian (GegĖ/Tosk) STEP10 v0.3

- Turkish STEP10 v0.1 (langHint=tr)

- Turkish STEP20 v0.1 (langHint=tr) + STEP10→STEP20 compare

Controls / drift visibility:

- Pseudowords STEP10/STEP20 v0.1 (negative control; no strengthening)

- Taiwan Zhuyin suite (fragility / weakening under expansion)

Results (copied from committed baselines)

Turkish (positive signal; strengthens)

- STEP10 presence mean: $r=-0.987$ ($p=0.000$), $\rho=-0.964$ ($p=0.002$)

- STEP20 presence mean: $r=-0.989$ ($p=0.000$), $\rho=-1.000$ ($p=0.000$)

- STEP10→STEP20 compare (presence mean): $\Delta|\rho|=+0.036$

Baselines:

- `tests/validation/baselines/turkish.spectrum.step10.v0.1.md|json`

- `tests/validation/baselines/turkish.spectrum.step20.v0.1.md|json`

- `tests/validation/baselines/turkish.spectrum.step20.v0.1.compare.v0.1.md|json`

Pseudowords (negative control; no strengthening)

- STEP10 presence mean: $r=-0.590$ ($p=0.166$), $\rho=-0.536$ ($p=0.239$)

- STEP20 presence mean: $r=-0.590$ ($p=0.158$), $\rho=-0.536$ ($p=0.227$)

- STEP10→STEP20 compare (presence mean): $\Delta|\rho|=0.000$

Baselines:

- `tests/validation/baselines/pseudowords.spectrum.step10.v0.1.md|json`

- `tests/validation/baselines/pseudowords.spectrum.step20.v0.1.md|json`

- `tests/validation/baselines/pseudowords.spectrum.step20.v0.1.compare.v0.1.md|json`

Taiwan Zhuyin (drift visibility)

Designed to show that some apparent signals **weaken** as N increases.

Baselines:

- `tests/validation/baselines/taiwan.spectrum.rootOnly.v1.0.compare.v0.1.md|json`
- `tests/validation/baselines/taiwan.spectrum.rootOnly.v1.0.audit.v0.1.md|json`
- `tests/validation/baselines/taiwan.spectrum.rootOnly.v1.0.toneSlope.v0.1.md|json`

Albanian (positive signal reference)

Baselines:

- `tests/validation/baselines/albanian.spectrum.gegTosk.step10.v0.3.md|json`

Reproduce (exact commands)

```bash

npm install

npm run gate:quick

npm test -- tests/research/turkish.spectrum.step20.v0.1.spec.ts

npm test -- tests/research/pseudowords.spectrum.step20.v0.1.spec.ts

npm test -- tests/research/taiwan.spectrum.rootOnly.v1.0.spec.ts